

## Assignment - 5 (Polymorphism)

que. List 20 polymorphism example.

1) Withdraw() on Bank account

Same msg: withdraw() is sent to all subclasses of Bank account.

- A Saving account : withdraw() { check mini. balance before deduct }
- B Current account : withdraw() { check overdraft limit -1- }
- C Loan account : withdraw() { check last transaction date before allowing }

result : implemented differently.

Same msg withdraw() sent to Bank account, but each subclass implements it's differently.

2) make sound() on animal

- A Dog : make sound() { print 'bark bark' }
- B cat : make sound() { print 'meow meow' }
- C cow : make sound() { print 'moo moo' }

3) get area() on shape

- A circle : get area() { return  $3.14 * \text{radius} * \text{radius}$  }
- B rectangle : get area() { return  $\text{length} * \text{width}$  }
- C Triangle : get area() { return  $0.5 * \text{base} * \text{height}$  }



4) calculateSalary() on employee

- A Manager: calculateSalary() { base salary + Team Bonus + allowance }
- B Developer: calculateSalary() { base salary + Project Bonus }
- C Intern: calculateSalary() { Fixed stipend only, no Bonus }

5) pay() on Payment

- A UPI Payment: pay() { validate UPI PIN & deduct from linked A/c }
- B Credit Card: pay() { Authorize card, deduct from credit limit }
- C Cash Payment: pay() { physically counting & hand over cash }

6) login() on User

- A Admin: login() { verify credentials + 2FA OTP required }
- B Customer: login() { verify username and password only }
- C Guest: login() { auto-login with limited access, no credentials }

7) move() on Vehicle

- A Car: move() { start engine, engage gear, press accelerator }
- B Bicycle: move() { pedal the wheels manually }
- C Train: move() { engage electric motor on railway track }



8) generateReport() on Department

- A Finance Dept: generateReport() { Product Profit/loss Balance }
- B HR Dept: generateReport() { Product employee attendance }
- C IT Dept: generateReport() { Product server uptime & bug tracking }

9)

9) SendNotification() on Notification service.

- A Email service: sendNotification() { compose & send via SMTP email }
- B SMS service: sendNotification() { send text msg via telecom gateway }
- C Push Notification: sendNotification() { send alert via Firebase to mobile app }

10) FileComplaint() on CustomerSupport

- A Product complaint: FileComplaint() { assign to product quality team + generate replacement order }
- B delivery complaint: FileComplaint() { assign to logistic team + track shipment }
- C Billing complaint: FileComplaint() { assign to finance team + review transaction }